

Heights I*

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1 Introduction (Faltings' theorem)

In arithmetic geometry, heights are gadgets that measure *complexity* or *size*, and they are useful for obtaining *finiteness theorems*.

Before formally introducing the various notions of *height* that we will use, let us mention how this shows up in Faltings' proof of Mordell's conjecture.

Mordell's conjecture over a number field F is the following statement:

$M(F)$ – For every smooth projective curve C over F of genus ≥ 2 , the set $C(F)$ is finite.

Parshin's argument reduces $M(F)$ to Shafarevich's conjecture. For fixed $g \in \mathbf{N}$, number field F and S a finite set of places of F , this conjecture states that:

$\text{Sha}(g, F, S)$ – There are only finitely many isomorphism classes of abelian varieties A over F of dimension $\dim A = g$ and with good reduction outside S .

Let k be a field, A be an abelian variety over k and $d \in \mathbf{N}$. Consider the following statements (**strong** and **weak hypothesis**):

$\text{Hyp}_s(k, A)$ – There are only finitely many isomorphism classes of abelian varieties k -isogenous to A .

$\text{Hyp}_w(k, A, d)$ – There are only finitely many isomorphism classes of abelian varieties k -isogenous to A and admitting a k -polarization of degree d^2 .

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For a fixed number field F , we have the following implications:

$$\text{Hyp}_w(F, A, d) \forall A \forall d \Rightarrow \text{Hyp}_s(F, A) \forall A \Rightarrow \text{Sha}(g, F, S) \forall g \forall S \Rightarrow M(F)$$

We recall that the first implication is given by Zarhin's trick and the second one is an argument of Faltings that combines Tate's conjecture and the semi-simplicity of the Galois action on Tate modules.

We have thus reduced Mordell's conjecture to proving the weak hypothesis for number fields. Faltings' idea is to use heights for obtaining this finiteness result. For fixed $g, d \in \mathbf{N}$ and field k of characteristic 0, let

$$\mathcal{M}_{g,d}(k) = \{(A, \lambda) \mid A \text{ abelian variety over } k, \lambda : A \rightarrow A^\vee \text{ polarization of degree } d^2\} / \sim$$

where $(A, \lambda) \sim (B, \mu)$ if there's an isomorphism $f : A \rightarrow B$ making the diagram

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \lambda \downarrow & & \downarrow \mu \\ A^\vee & \xleftarrow{f^\vee} & B^\vee \end{array}$$

One can show¹ that there is a $\mathbf{Q}$ -variety $M_{g,d}$ and a higher dimensional analog of the j -invariant

$$j : \mathcal{M}_{g,d}(k) \rightarrow M_{g,d}(k)$$

functorial in k and bijective when k is algebraically closed. The variety $M_{g,d}$ comes equipped with an immersion in $\mathbf{P}_{\mathbf{Q}}^n$, for some n , and thus we can define a height $h : M_{g,d}(\overline{\mathbf{Q}}) \rightarrow \mathbf{R}_{\geq 0}$. The **modular height** is just the composition with j :

$$\begin{aligned} h_M : \mathcal{M}_{g,d}(\overline{\mathbf{Q}}) &\rightarrow \mathbf{R}_{\geq 0} \\ (A, \lambda) &\mapsto h(j(A, \lambda)) \end{aligned}$$

The basic idea is to get the finiteness result from Northcott's theorem: if F is a number field and $X \in \mathbf{R}_{>0}$, then

$$\{x \in M_{g,d}(F) \mid h(x) \leq X\}$$

is finite. So far so good, but there are two difficulties with this approach:

1. For a fixed $j_0 \in M_{g,d}(k)$ and $(A_0, \lambda_0) \in \mathcal{M}_{g,d}(k)$ such that $j(A_0, \lambda_0) = j_0$, the set

$$\{(A, \lambda) \in \mathcal{M}_{g,d}(k) \mid j(A, \lambda) = j_0\}$$

consists of *twisted forms* of (A_0, λ_0) and is not finite in general.

Example 1.1. For the case of elliptic curves, we can take $d = g = 1$ and $M_{1,1} = \mathbf{A}_{\mathbf{Q}}^1$. Notice that two curves over a number field F can have the same j -invariant without being isomorphic over F . For instance, let $a \in F^\times \setminus F^{\times 2}$, $g_2, g_3 \in F$ such that $g_2^3 - 27g_3^2 \neq 0$, $E : y^2 = 4x^3 - g_2x - g_3$ and $E_a : y^2 = 4x^3 - a^2g_2x - a^3g_3$. Then $j(E) = j(E_a)$ but E is not F -isomorphic to E_a .

¹Certainly, not me...

This is bad because we want a finiteness result in $\mathcal{M}_{g,d}(F)$ and not in $M_{g,d}(F)$.

2. It is not clear how the modular height varies under isogenies, i.e. if A and B are isogenous, how do we compare $h_M(A, \lambda)$ and $h_M(B, \mu)$?

The first problem is easy to deal: we must impose a semi-stability hypothesis and then use the “semi-stable reduction theorem” of Grothendieck. Faltings’ idea (*toujours lui !*) to deal with the second one is to introduce *another* notion of height, nowadays called *Faltings height*. Namely, for every abelian variety A over a number field, Faltings attach a real number $h_{\text{Fal}}(A)$ by “measuring volumes”. He proves that the set of Faltings heights $h_{\text{Fal}}(A)$ of abelian varieties A having semi-stable reduction everywhere and isogenous to a fixed abelian variety A_0 is finite.

Finally, the only missing link is to compare the two heights h_{Fal} and h_M . In fact, Faltings proves that $h_{\text{Fal}}(A) = \alpha h_M(A, \lambda) + O(\log^+ h_M(A, \lambda) + 1)$, for some $\alpha \neq 0$, where $\log^+ t = \log t$ if $t \geq 1$ and is equal to 0 otherwise. This is really difficult and I will not touch it.

2 Arakelovian stuff

Adelic number theory teaches us that it is useful to consider all places of a number field in one unified object. In algebraic geometry, this same philosophy is realized by means of “compactifications”. For instance, for each point of $A^1(\mathbf{C}) \subset \mathbf{P}^1(\mathbf{C})$ there corresponds a valuation, given by an irreducible polynomial of $\mathbf{C}[X]$, and we also have an infinite valuation coming from $\infty \in \mathbf{P}^1(\mathbf{C})$. The analogy between number fields and function fields suggests that one should also be able to “compactify”, in some sense, the curve $\text{Spec } \mathcal{O}_F$, F being a number field. It is not clear how to do this geometrically, so Arakelov found an ad-hoc way of overcoming this difficulty.

2.1 Hermitian vector bundles on complex manifolds

Let M be a complex manifold (possibly non connected) and $E \rightarrow M$ a complex C^∞ -vector bundle. Recall that a **hermitian product** on E is a morphism of real vector bundles

$$h : E \times E \rightarrow \mathbf{C} \times M$$

such that, for each $p \in M$, the induced map

$$h_p : E_p \times E_p \rightarrow \mathbf{C}$$

is a hermitian product on the vector space E_p . Every hermitian product defines a norm² $\| \cdot \|_h : E \rightarrow \mathbf{R} \times M$ given at $p \in M$ by $\|v\|_{h,p} = \sqrt{h_p(v, v)}$, $v \in E_p$.

Remark 2.1. By a classical theorem, giving a hermitian product on a complex vector space is the same as giving a norm that satisfies the *parallelogram law*: $2\|v\|^2 + 2\|w\|^2 = \|v + w\|^2 + \|v - w\|^2$.

In particular, if E is a line bundle, giving a norm is the same thing as giving a hermitian product !

²A smooth map over M that is a norm at every fiber.

Example 2.1. Let $E = \mathbf{C} \times M$ be the trivial line bundle. Giving a hermitian product h on E is the same thing as giving a C^∞ function $\varphi : M \rightarrow \mathbf{R}$ by the rule $\varphi(p) = \log \|1\|_{h,p}$.

Definition 2.1. A complex vector bundle E with a hermitian product h is a **hermitian vector bundle** $\bar{E} = (E, h)$.

One can define what is an **exact sequence** of hermitian vector bundles over M : it is an exact sequence of the underlying bundles

$$0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$$

such that the norms of E' and E'' are induced by E , by restriction and quotient respectively. We say that $\bar{E}'$ is a sub-hermitian vector bundle of $\bar{E}$ and $\bar{E}''$ is the quotient $\bar{E}/\bar{E}'$. In particular we have a notion of isomorphism, although we do not define what is a morphism here.

One can also perform pullback of hermitian vector bundles and the usual multilinear constructions. Let us just give one example.

Example 2.2. Let (E, h) be a hermitian vector bundle and $E^\vee$ be the dual bundle of E . The function $\varphi : E \rightarrow E^\vee$ given at $p \in M$ by $\varphi_p(v) = h_p(-, v)$ is a semi-linear vector bundle isomorphism. We define the **dual** hermitian product $h^\vee$ on $E^\vee$ by

$$h_p^\vee(\varphi_p(v), \varphi_p(w)) = \overline{h_p(v, w)} = h_p(w, v)$$

One can verify, using Cauchy-Schwarz, that the norm $\| \cdot \|_{h^\vee}$ is just the usual dual norm of $\| \cdot \|_h$ in functional analysis.

Example 2.3 (Projective spaces). Let $M = \mathbf{P}^n(\mathbf{C}) = \{p \leq \mathbf{C}^{n+1} \mid \dim p = 1\}$ seen as a set of vector lines in $\mathbf{C}^{n+1}$. The tautological line bundle on M is

$$\mathcal{O}(-1) = \{(p, v) \in M \times \mathbf{C}^{n+1} \mid v \in p\}$$

The canonical hermitian product $\langle \cdot, \cdot \rangle$ on $\mathbf{C}^{n+1}$ induces a hermitian product on $\mathcal{O}(-1)$. Thus we have a natural hermitian product $h = \langle \cdot, \cdot \rangle^\vee$ on the dual line bundle $\mathcal{O}(-1)^\vee = \mathcal{O}(1)$:

$$h_p(\varphi, \psi) = \overline{\langle v, w \rangle}$$

where $\varphi = \langle -, v \rangle$, $\psi = \langle -, w \rangle$ and $v, w \in p$.³ One can easily verify that, if $u \in p$ satisfies $\langle u, u \rangle = 1$, then

$$h_p(\varphi, \psi) = \varphi(u) \overline{\psi(u)}$$

This is not the only “natural” hermitian product that we can consider on $\mathcal{O}(1)$. To get a feeling on how to find the other ones, consider for each i the global section

$$\begin{aligned} t^i : \mathbf{P}^n(\mathbf{C}) &\rightarrow \mathcal{O}(1) \\ p &\mapsto t_p^i \end{aligned}$$

where $t_p^i : p \rightarrow \mathbf{C}$ is given by $(v_0, \dots, v_n) \mapsto v_i$.

³This induces the so-called Fubini-Study metric on $\mathbf{P}^n(\mathbf{C})$. In fact, the real 2-form $\omega = \frac{1}{2i\pi} \partial\bar{\partial} \log h$ is Kähler and the Fubini-Study metric is the unique hermitian metric (i.e. hermitian product on the tangent bundle) associated to ω : $(v, w) \mapsto \omega(v, iw) - i\omega(v, w)$.

Remark 2.2. The section t^i trivializes $\mathcal{O}(1)$ over $U_i = \{(x_0 : \dots : x_n) \in \mathbf{P}^n(\mathbf{C}) \mid x_i \neq 0\}$. From the algebraic point of view, if we write $\mathbf{P}_{\mathbf{C}}^n = \text{Proj } \mathbf{C}[X_0, \dots, X_n]$, the t^i correspond to the monomials X_i .

The $t^0, \dots, t^n$ form a basis for $\Gamma(\mathbf{P}^n(\mathbf{C}), \mathcal{O}(1))$, thus to understand a hermitian product on $\mathcal{O}(1)$, it is sufficient to understand it composed on the t^i . If we write $p = (x_0 : \dots : x_n)$ and $u = \lambda^{-1}(x_0, \dots, x_n)$ where $\lambda = \sqrt{\sum_j |x_j|^2}$, then:

$$\|t_p^i\|_{h,p} = \sqrt{t_p^i(u) \overline{t_p^i(u)}} = \frac{|x_i|}{\lambda} = \frac{|x_i|}{\sqrt{\sum_j |x_j|^2}}$$

In general, for $q \in \{1, 2, 3, \dots, \infty\}$, the formula

$$\|t_p^i\|_{q,p} = \frac{|x_i|}{\sqrt[q]{\sum_j |x_j|^q}}$$

gives rise to the so-called L^q -norm on $\mathcal{O}(1)$. In particular, the norm associated to h is just the L^2 -norm.

2.2 Hermitian vector bundles on arithmetic varieties

Let F be a number field. We say that a morphism $X \rightarrow \text{Spec } \mathcal{O}_F$ is **generically smooth** if $X_F = X \otimes_{\mathcal{O}_F} F \rightarrow \text{Spec } F$ is smooth.

Definition 2.2. An **arithmetic variety** over F is a projective generically smooth morphism $X \rightarrow \text{Spec } \mathcal{O}_F$ such that X is regular and integral.

Some examples to keep in mind: $\text{Spec } \mathcal{O}_F$ and $\mathbf{P}_{\mathcal{O}_F}^n$ as arithmetic varieties over F .

Lemma 2.1. Let $S = \text{Spec } \mathcal{O}_F$. There's a canonical decomposition

$$S_{\mathbf{C}} = \text{Spec}(\mathcal{O}_F \otimes_{\mathbf{Z}} \mathbf{C}) = \bigsqcup_{\sigma} \text{Spec } \mathbf{C}$$

where $\sigma : F \rightarrow \mathbf{C}$ runs through the set of embeddings of F into $\mathbf{C}$.

This can be seen by number theory. Here's a proof via algebraic geometry.

Proof. We have the following commutative diagram with cartesian squares

$$\begin{array}{ccccc} S_{\mathbf{C}} & \longrightarrow & \text{Spec } F & \longrightarrow & S \\ \downarrow & & \downarrow & & \downarrow \\ \text{Spec } \mathbf{C} & \longrightarrow & \text{Spec } \mathbf{Q} & \longrightarrow & \text{Spec } \mathbf{Z} \end{array}$$

Since $\text{Spec } F \rightarrow \text{Spec } \mathbf{Q}$ is finite étale, $S_{\mathbf{C}} \rightarrow \text{Spec } \mathbf{C}$ is also finite étale and, since $\mathbf{C}$ is algebraically closed, we can write $A = \mathcal{O}_F \otimes_{\mathbf{Z}} \mathbf{C} = \prod_i A/\mathfrak{m}_i$ where the $\mathfrak{m}_i$ are the maximal ideals of A . It's easy to see that maximal ideals of A are in correspondence with embeddings $F \rightarrow \mathbf{C}$. ■

In particular, if $X \rightarrow \text{Spec } \mathcal{O}_F$ is an arithmetic variety, then $X_{\mathbf{C}} = X \otimes_{\mathbf{Z}} \mathbf{C}$ has a canonical decomposition

$$X_{\mathbf{C}} = \bigsqcup_{\sigma} X_{\sigma}$$

where $X_{\sigma} = X \times_{\mathcal{O}_F} \text{Spec } \mathbf{C}$ via the embedding $\sigma : F \rightarrow \mathbf{C}$.

Remark 2.3. In general, if $\tau \neq \sigma$, X_τ and X_σ are *not* isomorphic as $\mathbf{C}$ -varieties. Consider, for example, elliptic curves over a number field $F \neq \mathbf{Q}$.

We remark that $X(\mathbf{C}) = \bigsqcup_\sigma X_\sigma(\mathbf{C})$ can be given a natural structure of analytic space. The hypothesis made on X implies that $X(\mathbf{C})$ is actually a compact complex manifold, usually non connected.

For a coherent locally free sheaf E on X , we denote by $E(\mathbf{C})$ the induced vector bundle on $X(\mathbf{C})$. There is also a decomposition

$$E(\mathbf{C}) = \bigoplus_\sigma \sigma^* E(\mathbf{C})$$

where $\sigma^* E(\mathbf{C}) = E(\mathbf{C})|_{X_\sigma(\mathbf{C})}$.

Notice that $E(\mathbf{C})$ and $X(\mathbf{C})$ come equipped with canonical complex conjugations $v \mapsto \bar{v}$ and $p \mapsto \bar{p}$, given by composing a rational point $\text{Spec } \mathbf{C} \rightarrow X$ with the automorphism of $\mathbf{C}$ given by the complex conjugation. It induces bijections $X_\sigma(X) \rightarrow X_{\bar{\sigma}}(\mathbf{C})$ and $\sigma^* E(\mathbf{C}) \rightarrow \bar{\sigma}^* E(\mathbf{C})$.

Definition 2.3. A **hermitian vector bundle** on an arithmetic variety $X \rightarrow \text{Spec } \mathcal{O}_F$ is a couple $\bar{E} = (E, h)$ where E is a coherent locally free sheaf on X and h is a hermitian product on $E(\mathbf{C})$ that is

- (i) orthogonal for the decomposition $E(\mathbf{C}) = \bigoplus_\sigma \sigma^* E(\mathbf{C})$
- (ii) invariant under complex conjugation: $h_{\bar{p}}(\bar{v}, \bar{w}) = \overline{h_p(v, w)}$ for every $p \in X(\mathbf{C})$ and $v, w \in E(\mathbf{C})_p$.

We say that a hermitian vector bundle $\bar{E}$ is a **hermitian line bundle** if E is invertible.

Remark 2.4. To give a hermitian product h on $E(\mathbf{C})$ satisfying (i) and (ii) is the same as to give a collection of hermitian products h_σ on $\sigma^* E(\mathbf{C})$ such that

$$h_{\bar{\sigma}, \bar{p}}(\bar{v}, \bar{w}) = \overline{h_{\sigma, p}(v, w)}$$

for every embedding $\sigma : F \rightarrow \mathbf{C}$, $p \in X_\sigma(\mathbf{C})$ and $v, w \in \sigma^* E(\mathbf{C})_p$.

Again, we can define exact sequences, multilinear constructions and pullback of hermitian vector bundles on arithmetic varieties. If F'/F is an extension of number fields, $X \rightarrow \text{Spec } \mathcal{O}_F$ is an arithmetic variety, and $\bar{E}$ is a hermitian vector bundle on X , there is also a notion of base change: we can define in a natural way a structure of hermitian vector bundle on the coherent locally free sheaf $E \otimes_{\mathcal{O}_F} \mathcal{O}_{F'}$ over the arithmetic variety $X \otimes_{\mathcal{O}_F} \mathcal{O}_{F'} \rightarrow \text{Spec } \mathcal{O}_{F'}$.

Example 2.4. By the examples of the previous section, there are natural L^p hermitian structures (for $p = 1, \dots, \infty$) on the sheaf $\mathcal{O}(1)$ over $\mathbf{P}_{\mathcal{O}_F}^n$.

Example 2.5. Let F be a number field and $S = \text{Spec } \mathcal{O}_F$. Consider the trivial line bundle $L = \mathcal{O}_S$ on S . Let us describe the hermitian products on L . Let $\sigma_1, \dots, \sigma_{r_1}$ be the real embeddings of F and $\sigma_{r_1+1}, \bar{\sigma}_{r_1+1}, \dots, \sigma_{r_1+r_2}, \bar{\sigma}_{r_1+r_2}$ be the complex ones.

For a tuple $(u_i) \in \mathbf{R}^{r_1+r_2}$, we put

$$h_{\sigma_i} : \mathbf{C} \times \mathbf{C} \rightarrow \mathbf{C}$$

$$(z, w) \mapsto e^{-u_i} z \bar{w}$$

and $h_{\bar{\sigma}_i} = h_{\sigma_i}$ for $r_1 + 1 \leq i \leq r_1 + r_2$. Then it is clear that the collection (h_σ) induces a hermitian product on L and that every hermitian product on L is of this form.

2.3 Heights

If we consider $S = \text{Spec } \mathcal{O}_F$ as an arithmetic variety over F , the situation is particularly nice, since a hermitian vector bundle over S corresponds to a projective module of finite rank M over $\mathcal{O}_F$ equipped with hermitian products h_σ on $\sigma^*M = M \otimes_\sigma \mathbb{C}$ invariant under complex conjugation. We denote by $\| \cdot \|_\sigma$ the norm defined by h_σ .

Definition 2.4. We call $\bar{M} = (M, (h_\sigma))$ a **metrized module** over $\mathcal{O}_F$.

Metrized modules of rank 1 are also called “metrized line bundles”. Let $\bar{L}$ be a metrized line bundle over $\mathcal{O}_F$ and $s \in L \setminus \{0\}$.

Definition 2.5. The (Arakelov) **degree** of $\bar{L}$ is denoted and defined by

$$\widehat{\deg} \bar{L} = \log |L/\mathcal{O}_F s| - \sum_{\sigma} \log \|s\|_{\sigma}$$

We remark that $s \neq 0$ implies that $\|s\|_{\sigma} \neq 0$, so $\log \|s\|_{\sigma}$ is well defined.

Lemma 2.2. The degree of $\bar{L}$ does not depend on the choice of s .

Let $d = [F : \mathbb{Q}]$.

Proof. Let $t \in L \setminus \{0\}$. Since $L \otimes F$ is one-dimensional over F , there exists $\alpha \in F^\times$ such that $t \otimes 1 = s \otimes \alpha$. By writing $\alpha = a/b$, with $a, b \in \mathcal{O}_F \setminus \{0\}$ and using the injectivity of $L \rightarrow L \otimes F$ we see that $as = bt$. In particular, we can suppose that $\alpha \in \mathcal{O}_F \setminus \{0\}$ and $t = \alpha s$.

Since the support of $L/\mathcal{O}_F t$ is closed and different from $\text{Spec } \mathcal{O}_F$ (because $\mathcal{O}_F t \otimes F = L \otimes F$), it is finite.⁴ In particular,

$$L/\mathcal{O}_F t \cong \bigoplus_{\mathfrak{p}} L_{\mathfrak{p}}/\mathcal{O}_{F,\mathfrak{p}} t$$

where $\mathfrak{p}$ runs through the set of maximal ideals of $\mathcal{O}_F$, and we get

$$\begin{aligned} \log |L/\mathcal{O}_F t| &= \sum_{\mathfrak{p}} \log |L_{\mathfrak{p}}/\mathcal{O}_{F,\mathfrak{p}} t| \\ &= \sum_{\mathfrak{p}} \log |L_{\mathfrak{p}}/\mathcal{O}_{F,\mathfrak{p}} \alpha s| \\ &= \sum_{\mathfrak{p}} (\log |L_{\mathfrak{p}}/\mathcal{O}_{F,\mathfrak{p}} s| + \log |\mathcal{O}_{F,\mathfrak{p}}/\alpha \mathcal{O}_{F,\mathfrak{p}}|) \\ &= \sum_{\mathfrak{p}} \log |L_{\mathfrak{p}}/\mathcal{O}_{F,\mathfrak{p}} s| - \sum_{\mathfrak{p}} \log |\mathcal{O}_{F,\mathfrak{p}}/\mathfrak{p}|^{-\nu_{\mathfrak{p}}(\alpha)} \end{aligned}$$

On the other hand, we have $\|t\|_{\sigma} = |\sigma(\alpha)| \|s\|_{\sigma}$. Thus

$$\begin{aligned} \log |L/\mathcal{O}_F t| - \sum_{\sigma} \log \|t\|_{\sigma} &= \sum_{\mathfrak{p}} \log |L_{\mathfrak{p}}/\mathcal{O}_{F,\mathfrak{p}} s| - \sum_{\sigma} \log \|s\|_{\sigma} - \sum_{\mathfrak{p}} \log |\mathcal{O}_{F,\mathfrak{p}}/\mathfrak{p}|^{-\nu_{\mathfrak{p}}(\alpha)} - \sum_{\sigma} \log |\sigma(\alpha)| \\ &= \log |L/\mathcal{O}_F s| - \sum_{\sigma} \log \|s\|_{\sigma} \end{aligned}$$

by the product formula. ■

Example 2.6. If $L = \alpha \mathcal{O}_F$ is a principal fractional ideal of $\mathcal{O}_F$, then we can take $s = \alpha$ and $\widehat{\deg} \bar{L} = -\sum_{\sigma} \log \|\alpha\|_{\sigma}$.

⁴The support of a coherent sheaf over a noetherian scheme X is always closed. If X is noetherian and integral of dimension 1, then closed sets are either X or finite sets of closed points.

Now, let $X \rightarrow \operatorname{Spec} \mathcal{O}_F$ be an arithmetic variety and $\bar{L}$ a hermitian line bundle on X .

Definition 2.6. The **height** associated to $\bar{L}$ is the function:

$$h_{\bar{L}} : X(\mathcal{O}_F) \rightarrow \mathbf{R}$$

$$x \mapsto \frac{1}{[F : \mathbf{Q}]} \widehat{\deg} x^* \bar{L}$$

Notice that, since $X \rightarrow \operatorname{Spec} \mathcal{O}_F$ is projective and $\mathcal{O}_F$ is a Dedekind domain, the natural map $X(\mathcal{O}_F) \rightarrow X(F)$ is bijective.⁵ Thus, we have a height function $h_{\bar{L}} : X(F) \rightarrow \mathbf{R}$. In fact, by base change (cf. the exercise below), we get a well define height

$$h_{\bar{L}} : X(\bar{F}) \rightarrow \mathbf{R}$$

Exercise. Let F'/F be an extension of number fields, $X' = X \otimes_{\mathcal{O}_F} \mathcal{O}_{F'} \rightarrow \operatorname{Spec} \mathcal{O}_{F'}$ the base change of $X \rightarrow \operatorname{Spec} \mathcal{O}_F$ and $L' = L \otimes_{\mathcal{O}_F} \mathcal{O}_{F'}$ with the pullback hermitian structure. Show that, if $x \in X'(\mathcal{O}_{F'}) = X(\mathcal{O}_F)$, then $h_{\bar{L}'}(x) = h_{\bar{L}}(x)$.

Let us describe the bijection $X(\mathcal{O}_F) \rightarrow X(F)$ more explicitly for $X = \mathbf{P}_{\mathcal{O}_F}^n$.

Recall that, if S is a scheme, to give a morphism $S \rightarrow \mathbf{P}^n$ is the same as to give an equivalence class of a line bundle L on S with a sequence of $n+1$ global sections s_i that generate L , where two such gadgets $(L, (s_i))$ and $(L', (s'_i))$ are equivalent if there is a line bundle isomorphism $\varphi : L \rightarrow L'$ such that $\varphi^*(s'_i) = s_i$ for every i . Now, if $S = \operatorname{Spec} A$ where A is a Dedekind ring with fraction field K , then there is a canonical group isomorphism $\operatorname{Cl}(A) \rightarrow \operatorname{Pic}(S)$. Thus, giving a $(L, (s_i))$ up to equivalence amounts to giving a fractional ideal $\mathfrak{a} \subset A$ with a sequence of $n+1$ generating elements x_i , up to scaling by an element of $K^\times$.

Remark 2.5. Notice that $\mathfrak{a}$ is “superfluous”, since $\mathfrak{a} = \sum_i A x_i$. Thus, to give a morphism $S \rightarrow \mathbf{P}^n$ is the same thing as to give a point $(x_0 : \dots : x_n) \in \mathbf{P}^n(K)$.

Now, the main concrete example:

Example 2.7. Consider $\mathbf{P}_{\mathcal{O}_F}^n$ as an arithmetic scheme over F and $\overline{\mathcal{O}(1)}$ with the L^p norm ($p = 1, \dots, \infty$). We denote h_{L^p} the associated height. Let us compute it explicitly. Let $x = (x_0, \dots, x_n) \in \mathbf{P}^n(F)$ and $x : \operatorname{Spec} \mathcal{O}_F \rightarrow \mathbf{P}^n$ the corresponding morphism. Up to isomorphism, $x^* \mathcal{O}_F$ is induced by the fractional ideal

$$\mathfrak{a} = \mathcal{O}_F x_0 + \dots + \mathcal{O}_F x_n$$

and $x^*(X_i) = x_i$. Notice that

$$\|x_0\|_\sigma = \frac{|t^0(\sigma(x))|}{\sqrt[p]{\sum_i |t^i(\sigma(x))|^p}} = \frac{|\sigma(x_0)|}{\sqrt[p]{\sum_i |\sigma(x_i)|^p}}$$

Thus:

$$\begin{aligned} h_{L^p}(x) &= \frac{1}{[F : \mathbf{Q}]} \left(\log |\mathfrak{a} / \mathcal{O}_F x_0| - \sum_\sigma \log \|x_0\|_\sigma \right) \\ &= \frac{1}{[F : \mathbf{Q}]} \left(\left(\sum_{\mathfrak{p}} (\nu_{\mathfrak{p}}(\mathfrak{a}) - \nu_{\mathfrak{p}}(x_0) \log |\mathcal{O}_F / \mathfrak{p}|) \right) - \left(\sum_\sigma (\log |\sigma(x_0)| - \log (\sum_i |\sigma(x_i)|^p)^{1/p}) \right) \right) \\ &= \frac{1}{[F : \mathbf{Q}]} \left(\sum_{\mathfrak{p}} \nu_{\mathfrak{p}}(\mathfrak{a}) \log |\mathcal{O}_F / \mathfrak{p}| + \sum_\sigma \log (\sum_i |\sigma(x_i)|^p)^{1/p} \right) \end{aligned}$$

⁵Actually, as Yang points out, this is true more generally for a proper morphism, by a combination of the valuative criterion of properness and the “smearing lemma” (EGA I 6.5).

Since $\nu_p(a) = \min_i \nu_p(x_i)$, we obtain

$$h_{L^p}(x) = \sum_{\nu \nmid \infty} \log \max_i |x_i|_\nu + \frac{1}{[F:\mathbf{Q}]} \sum_{\sigma} \log \left(\sum_i |\sigma(x_i)|^p \right)^{1/p}$$

In particular,

$$\begin{aligned} h_{L^\infty}(x) &= \sum_{\nu \nmid \infty} \log \max_i |x_i|_\nu + \frac{1}{[F:\mathbf{Q}]} \sum_{\sigma} \log \max_i |\sigma(x_i)| \\ &= \sum_{\nu} \log \max_i |x_i|_\nu \\ &= \log \prod_{\nu} \max_i |x_i|_\nu \end{aligned}$$

where ν runs through the set of places of F and, for $x \in F_\nu$ (the completion of F with respect to ν) and $\nu \nmid p$,

$$|x|_\nu = |N_{F_\nu/\mathbf{Q}_p} x|^{1/[F:\mathbf{Q}]}$$

is the normalized absolute value à la Serre (cf. [2]).

Definition 2.7. The height $h_{L^\infty} = h : \mathbf{P}^n(\overline{\mathbf{Q}}) \rightarrow \mathbf{R}$ is the (logarithmic) **Weil height**.

3 Northcott's theorem

If $x \in \mathbf{P}^n(\overline{\mathbf{Q}})$, we define the **degree** of x by

$$\deg x = \min\{[F:\mathbf{Q}] \mid F \text{ number field such that } x \in \mathbf{P}^n(F)\}$$

Equivalently, if $x = (x_0 : \cdots : x_n)$, $\deg x$ is the degree of the field obtained by adjoining x_i/x_j for all i, j ($x_j \neq 0$) to $\mathbf{Q}$.

Let $h : \mathbf{P}^n(\overline{\mathbf{Q}}) \rightarrow \mathbf{R}$ be the Weil height. By the product formula, it is easy to see that $h(x) \geq 0$ for all $x \in \mathbf{P}^n(\overline{\mathbf{Q}})$.

Theorem 3.1 (Northcott). For $X \in \mathbf{R}_{>0}$ and $d \in \mathbf{N}$, the set

$$\{x \in \mathbf{P}^n(\overline{\mathbf{Q}}) \mid h(x) \leq X \text{ and } \deg x \leq d\}$$

is finite.

In particular, for every number field F , $\{x \in \mathbf{P}^n(F) \mid h(x) \leq X\}$ is finite.

Proof of the theorem. It is sufficient to prove that the sets

$$\{x \in \mathbf{P}^n(\overline{\mathbf{Q}}) \mid h(x) \leq X \text{ and } \deg x = d\}$$

are finite.

This is easy for $d = 1$, since any $x \in \mathbf{P}^n(\mathbf{Q})$ can be written uniquely, up to sign, as $x = (a_0 : \cdots : a_n)$, for $a_i \in \mathbf{Z}$ having non common factor, and $h(x) = \log \max_i |a_i|_\infty$ (exercise!).

Now, let $x \in \mathbf{P}^n(\overline{\mathbf{Q}})$ with $\deg x = d$, F be the unique number field of degree d such that $x \in \mathbf{P}^n(F)$ and $\{\sigma_1, \dots, \sigma_d\}$ be the set of embeddings $F \rightarrow \overline{\mathbf{Q}}$. Write $x = (x_0 : \cdots : x_n)$ with $x_i \in F$. We put

$$x^i = (\sigma_i(x_0) : \cdots : \sigma_i(x_n)) = (x_0^i : \cdots : x_n^i)$$

Let S be the d -fold symmetric product of $\mathbf{P}^n$, i.e. the scheme obtained by quotienting $\prod_{i=1}^d \mathbf{P}^n$ by the diagonal action of the symmetric group $\mathfrak{S}_d$. We have a natural diagonal map $f_{\mathbb{F}} : \mathbf{P}^n(\mathbb{F}) \rightarrow \Sigma(\overline{\mathbf{Q}})$ given by $x \mapsto (x^1, \dots, x^d) \bmod \mathfrak{S}_d$. Since $f_{\mathbb{F}}(x)$ is fixed by $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$, we get $f_{\mathbb{F}}(x) \in \Sigma(\mathbf{Q})$.

Since $\{x \in \mathbf{P}^n(\overline{\mathbf{Q}}) \mid \deg x = d\} = \bigsqcup_{[\mathbb{F}:\mathbf{Q}]=d} \{x \in \mathbf{P}^n(\mathbb{F}) \mid \deg x = d\}$, we get a well define function

$$f : \{x \in \mathbf{P}^n(\overline{\mathbf{Q}}) \mid \deg x = d\} \rightarrow \Sigma(\mathbf{Q})$$

with *finite fibers*.

Let V be the $\mathbf{Q}$ -vector space consisting of homogeneous polynomials of degree d in $n+1$ variables and $g : \Sigma(\mathbf{Q}) \rightarrow \mathbf{P}(V)$ be the map induced by

$$(x^1, \dots, x^n) \mapsto (x_0^1 T_0 + \dots + x_n^1 T_n) \cdots (x_0^d T_0 + \dots + x_n^d T_n)$$

Notice that g is injective since $\mathbf{Q}[T_0, \dots, T_n]$ is an UFD. Let $\varphi = g \circ f$. Since conjugate elements have same absolute values, one easily checks that there exists some $C \in \mathbf{R}_{>0}$ such that

$$h(\varphi(x)) \leq C + dh(x)$$

Now, $\mathbf{P}(V) = \mathbf{P}^N(\mathbf{Q})$ for some N , thus $\{y \in \mathbf{P}(V) \mid h(y) \leq C + dX\}$ is finite. In particular, $\{\varphi(x) \mid x \in \mathbf{P}^n(\overline{\mathbf{Q}}), \deg(x) = d, h(x) \leq X\}$ is finite and, since the fibers of φ are finite, $\{x \in \mathbf{P}^n(\overline{\mathbf{Q}}) \mid \deg x = d, h(x) \leq X\}$ is finite. $\blacksquare$

Remark 3.2. This proof became uglier than I thought. It was adapted from the proof in Serre's book [2], 2.4. Strangely enough, Serre's proof seems to work only for proving that $\{x \in \mathbf{P}^n(\mathbb{F}) \mid h(x) \leq X\}$ is finite for every $\mathbb{F}$, which is often the way that Northcott's theorem is used but not (a priori) quite the same thing as the full statement.

To finish, let us mention that heights are also nice for “detecting torsion points” (cf. Néron-Tate height). We can illustrate this general principle with the following corollary.

Consider the embedding

$$\mathbf{G}_m = \text{Spec } \mathbf{Z}[X, Y]/(XY - 1) \rightarrow \mathbf{P}^1$$

given by $x \mapsto (1 : x)$ on rational points. Thus, there is defined a height function $h : \mathbf{G}_m(\overline{\mathbf{Q}}) = \overline{\mathbf{Q}}^\times \rightarrow \mathbf{R}_{\geq 0}$ and one easily check that $h(x^n) = nh(x)$ for every $x \in \overline{\mathbf{Q}}^\times$ and $n \in \mathbf{N}$.

Corollary 3.1. Let $x \in \overline{\mathbf{Q}}^\times$. Then, x is a root of unity if and only if $h(x) = 0$.

Proof. Let $\mathbb{F}$ be a number field containing x . If $x^n = 1$, then $|x|_v = 1$ for every place v of $\mathbb{F}$, thus $h(x) = \log \prod_v \max\{1, |x|_v\} = \log 1 = 0$. Conversely, if $h(x) = 0$, then $h(x^n) = nh(x) = 0$ for every $n \in \mathbf{N}$. By Northcott's theorem, the set $\{1, x, x^2, x^3, \dots\}$ must be finite, thus x is a root of unity. $\blacksquare$

We get for free the following classical fact:

Corollary 3.2 (Kronecker's theorem). Let α be an algebraic integer such that $|\sigma\alpha| = 1$ for every embedding $\sigma : \overline{\mathbf{Q}} \rightarrow \mathbf{C}$, then α is a root of unity.

References

- [1] Milne, J. S., *Abelian varieties*. Version 2.0. Available online at <http://www.jmilne.org/math/CourseNotes/AV.pdf>.
- [2] Serre, J-P., *Lectures on the Mordell-Weil theorem*. Aspects of Mathematics. Vieweg, 1987.